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LESIONS OF "ENCEPHALOMYELITIS" AND MULTIPLE SCLEROSIS

VENOUS THROMBOSIS AS THE PRIMARY ALTERATION

TRACY J. PUTNAM, M.D.

BOSTON

The experimental production in animals of venular thrombi scattered throughout the body has been reported by Flexner,¹ Kusama² and others following the administration of a great variety of substances. These include homologous and heterologous serum, organ extracts, bacterial filtrates, bacterial suspensions, colloidal materials, fat solvents, and salts of some heavy metals. The thrombi are of various types, some consisting chiefly of agglutinated platelets, others chiefly of stromas, others of strands of fibrin or amorphous material. Evidence has been submitted by Kusama² to show that these thrombi may redissolve in the course of time without becoming organized. Hoefer,³ working at the Neurological Unit, is in the process of repeating the fundamental observations and finds that of the various fractions of heterologous plasma the globulin is most effective in precipitating intravascular clotting (fig. 1), and that of the various organ extracts lung extract of Mills⁴ is perhaps the most potent. Campbell and Alexander⁵ have produced amorphous capillary and venular thrombi in the brain in experimental local anaphylaxis.

Similar thrombi in both fresh and advanced stages are frequently observed in human pathologic studies, although surprisingly little attention has been paid to them. I have been unable to find any systematic study of them, although they are familiar to every pathologist. They constitute a problem which is entirely different from that of thrombosis in large vessels, such as the heart or aorta, since the flow of blood stops entirely in the venule or capillary once the clot forms. They are therefore apt to lead to necrosis of the vessel wall rather than to organization and recanalization.

Venular thrombi occur particularly in the course of infectious diseases (fig. 2) and indeed are such a commonplace finding that they often escape comment. They have been more thoroughly studied in connection

with certain types of nephritis⁶ and appear to provoke a chronic inflammatory reaction in the kidney. They have also been reported in late death after carbon monoxide poisoning,⁷ and I have found them in a case of nitrous oxide poisoning.

Little attention has been paid to the occurrence of such thrombi in the brain, although it might well be expected that more serious damage should be produced by vascular obstruction there than in any other organ. In the earlier experimental work convulsions, coma and paralyzes were regularly reported, but no histologic examination of the nervous system was given. Thrombosis of veins of the white matter, producing demyelination, was reported by Meyer⁷ and others in a carbon monoxide poisoning. Hoefer's³ and Campbell and Alexander's⁵ experiments were directed toward a study of the cerebral lesions, and they show that clots are as easily provoked in the central nervous system as in any other organ. Indeed, there is some reason to believe that the tortuosity of the cerebral veins and the slowness of the flow within them predispose them to thrombosis.

THE RESULTS OF VENOUS OBSTRUCTION IN THE BRAIN

The effects of venous obstruction in the central nervous system have been studied experimentally⁸ and pathologically in this laboratory. The two sets of observations lead to the same conclusions. The stasis affects the white matter more than the gray. The commonest lesion consists of a swelling and disintegration of myelin upstream along the vein (fig. 3), accompanied in the severer lesions by some swelling and fragmentation of axis cylinders. There is a diffuse proliferation of astrocytes in the vicinity (fig. 4). At a later stage, phagocytes containing products of parenchymal destruction accumulate within the adventitial sheaths of vessels. Occasionally cells resembling lymphocytes are found.

This histologic picture, characterized by the predominance of acute exudative phenomena, changes gradually to one in which cellular activity is almost at a standstill. At most a few phagocytes are found within hypertrophied adventitial meshes. The axis cylinders resume a more normal appearance, and it often becomes difficult to be sure that any are missing. The destroyed myelin is apparently never regenerated, and the rough, indistinct edges of the area of myelin loss gradually become smooth and distinct. The glial fibrosis continues to grow more dense at least up to a year after the insult has occurred.

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From the Neurological Unit, Boston City Hospital, and the Department of Neurology, Harvard Medical School.

This is the thirteenth of a series of papers based on investigations the expenses of which were defrayed in part by the Multiple Sclerosis Fund of Harvard University.

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THE LESIONS OF VENOUS THROMBOSIS AND
THOSE OF "ENCEPHALITIS"

Neuropathologists will recognize at once that the acute lesion which has been described corresponds in all its details with that which is considered characteristic of certain types of "encephalomyelitis"; for example, the postvaccinal and postmeasles forms.⁹ In these con-

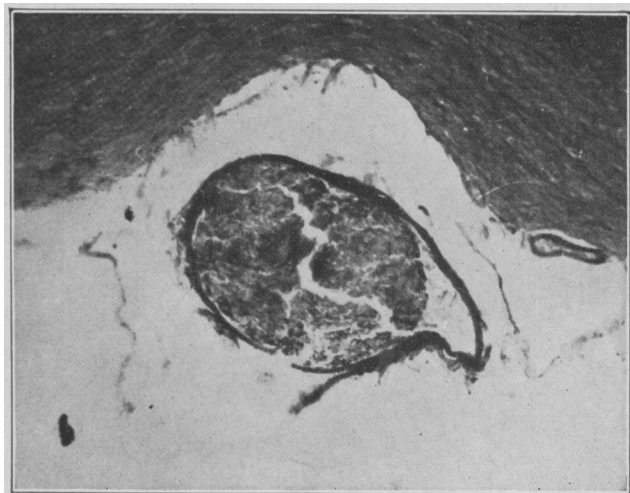


Fig. 1.—Clot in a cerebral venule of a rabbit that died forty minutes after the intravenous injection of lung extract. The clot consists of red cells caught in a dense network of curved strands of fibrin. Mallory's connective tissue stain, low power.

ditions congestion and thrombosis are almost regularly observed (figs. 5 and 6).¹⁰ Thrombi were present in other organs also in the only case of this type in which I have been able to study them (fig. 2).

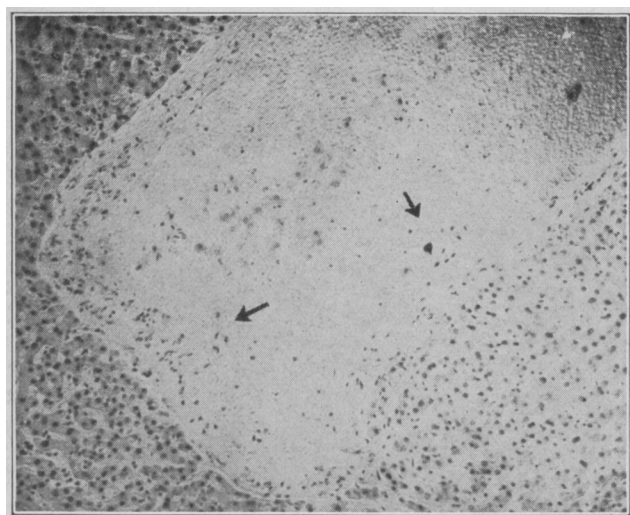


Fig. 2.—Thrombus from the liver of a patient dying from an "encephalitis" following tonsillitis. Arrows indicate invasion by endothelial cells. Mallory's connective tissue stain, low power.

The possibility that postinfectious "encephalomyelitis" is an allergic phenomenon has been suggested by Glanzmann,¹¹ van Bogaert¹² and others. The onset of the cerebral symptoms coincides with the development of immunity in a striking manner. The relation of allergy to thrombosis and of thrombosis to the cerebral

changes places the entire sequence in a clearer light. It also becomes easier to understand why these "encephalitides" occasionally occur in a hemorrhagic form and how they are related to the cerebral lesions produced by toxic substances such as arsphenamine, carbon monoxide and nitrous oxide.

THE RELATION OF VENOUS THROMBOSIS TO
MULTIPLE SCLEROSIS

It has long since been pointed out that lesions closely resembling those of the postinfectious "encephalomyelitides" may occur in the course of multiple sclerosis, and a careful comparison of various aspects of the two

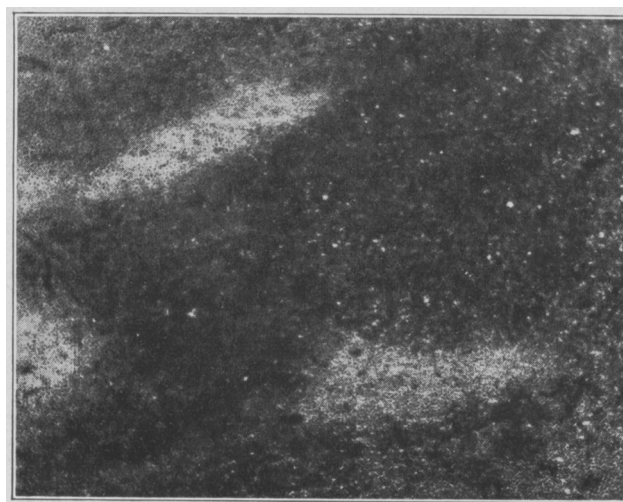


Fig. 3.—Local destruction of myelin fourteen days after experimental obstruction of cerebral venules in a dog. Mallory's connective tissue stain, low power.

groups shows that no sharp dividing line can be drawn between them.¹³ In both conditions, myelin is more damaged than the axis cylinders and can never regenerate in the central nervous system. The axis cylinders are injured or sometimes entirely destroyed in both dis-

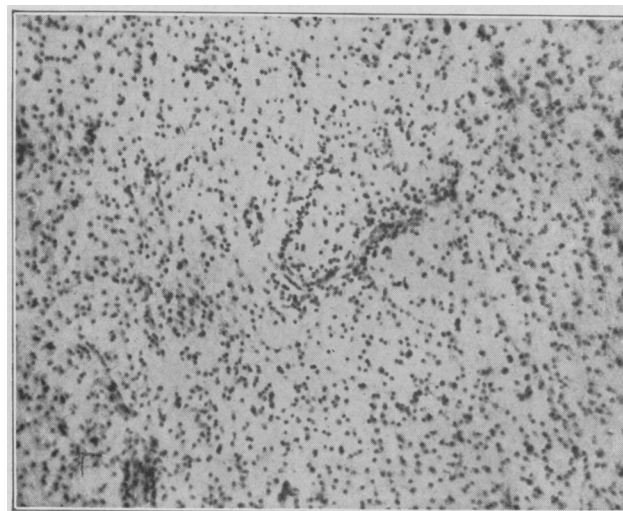


Fig. 4.—Diffuse glial proliferation in the cerebral white matter fourteen days after experimental obstruction of cerebral venules in a dog. Section from the same block as shown in figure 3. Gallocyanin stain, low power.

eases. The intensity of destruction of axis cylinders is greater the more severe and abrupt is the onset of the lesion, and it is therefore usually greater in fatal

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cases of "encephalitis" than in some of the mild, chronic lesions of multiple sclerosis. In the majority of sclerotic plaques, however, many axis cylinders are destroyed.¹⁴ A diffuse glial reaction occurs in areas of fresh degeneration in cases of multiple sclerosis, which is entirely comparable to that seen in "encephalomyelitis." At a later stage in both diseases phagocytic cells of various types accumulate about vessels and later still disappear. The glial scarring that gives multiple sclerosis its name is a slow, chronic process which takes place in much the same way in practically all diseases accompanied by loss of myelin; for example, old arteriosclerotic lesions¹⁵ or secondary degeneration.¹⁴ The typical sclerotic plaque is to be regarded, therefore, as a late stage of the lesion of a certain type of "encephalomyelitis."

The theory of the vascular origin of the lesions of multiple sclerosis has recently received support from two unexpected sources. In the first place, Alexander

sent phagocytized blood pigment rather than fragments of spirochetes, as suggested by Steiner.

This point of view is entirely consistent with the intermittent course of remissions and exacerbations



Fig. 5.—Thrombus in a cerebral vein from a case of postvaccinal "encephalitis." Masses of fibrin enclose red cells. Mallory's connective tissue stain, low power. Compare with figure 1.

and Myerson¹⁶ have shown by means of micro-incineration that a local demineralization is a characteristic feature both of anemic foci and of sclerotic plaques, in sharp contrast to the hypermineralization of inflammatory lesions.

In the second place, Blackman¹⁷ has recently repeated Steiner's work¹⁸ on the occurrence of certain cells which take a specific impregnation with silver in multiple sclerosis and dementia paralytica. Blackman finds that these typical cells occur also in areas of cerebral thrombosis, if the patient has survived a week or more (fig. 7). It appears, therefore, that they repre-

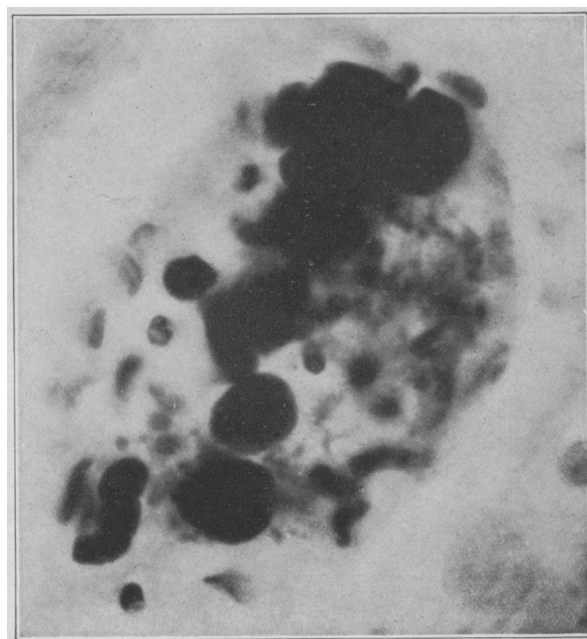


Fig. 6.—Thrombus in a cerebral vein from a case of "encephalitis" following antirabic inoculation. The dark masses apparently represent fibrin. Masson stain, oil immersion lens.

characteristic of multiple sclerosis, and with what is known of the factors precipitating attacks of the disease. In an extensive review of the subject von



Fig. 7.—Typical "silver cells" about a vessel, from a case of cerebellar thrombosis in which the lesion was approximately two weeks old. Steiner's stain, oil immersion.

Hoesslin¹⁹ lists the exogenous circumstances directly preceding exacerbation in 516 cases as follows: injury and accident, 11.4 per cent; pregnancy and labor, 5.8

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per cent; febrile diseases, 4.2 per cent; overexertion, 3.9 per cent; chilling, 2.3 per cent; fright and emotion, 7.0 per cent, and poisoning, 0.5 per cent.

It is striking that these factors are exactly those which predispose to disturbances of the stability of the blood plasma and to thrombosis.²⁰ Simon and Solomon²¹ have shown that the blood of patients with multiple

and 9). Thrombi also occur in other organs in cases of multiple sclerosis, usually without causing lesions.²² Further reasons have been given elsewhere for considering that the thrombi are primary to the other changes in the lesion and not secondary.²²

SUMMARY

Many of the factors that may lead to a diffuse thrombosis of small veins throughout the body correspond with the recognized antecedents of certain types of "encephalomyelitis" and the onset or exacerbation of some cases of multiple sclerosis. In both these groups of disease, thrombi and other evidences of vascular occlusion are found in relation to the characteristic lesions and in other organs of the body. These lesions correspond closely to the results of experimental and spontaneous occlusion of veins.

It seems probable, therefore, that the primary alteration in certain types of "encephalomyelitis" and multiple sclerosis is an abnormal lability of the blood plasma, which may be caused to clot in venules of the brain (and other organs) by a variety of exogenous (and probably also endogenous) factors.

MECKEL'S DIVERTICULUM

ASSOCIATED WITH INTUSSUSCEPTION AND ADENOCARCINOMA OF ECTOPIC GASTRIC MUCOSA:
REPORT OF CASE

HOWARD K. GRAY, M.D.

AND

JAMES W. KERNOHAN, M.D.

ROCHESTER, MINN.

That there is a persistence of the proximal end of the yolk stalk to form a pouch at its point of union with the intestine in approximately 3 per cent of all individuals has been widely recognized. This pouch is usually referred to as "Meckel's diverticulum" and is a structure of clinical importance because of the rôle it may play in producing intestinal obstruction. Benign ulcerating processes in ectopic gastric mucosa have been described with relative frequency and have been incriminated occasionally as the source of intestinal hemorrhage. The development of malignant lesions in the ectopic gastric mucosa of a Meckel's diverticulum is extremely rare, and for this reason we felt that it was justifiable to report such an incident. So far as we have been able to determine, carcinoma of Meckel's diverticulum has been reported in a limited number of cases. According to Franke,¹ Jores noted a primary carcinoma of the cylindric epithelium of a Meckel's diverticulum and reported this in 1899. Oberndorfer² cited one case in 1907, and other instances of this condition have been noted by Schaetz³ (1925), and by Hertzog and Carlson⁴ (1935).

From the Division of Surgery (Dr. Gray) and the Section on Pathologic Anatomy (Dr. Kernohan), the Mayo Clinic.

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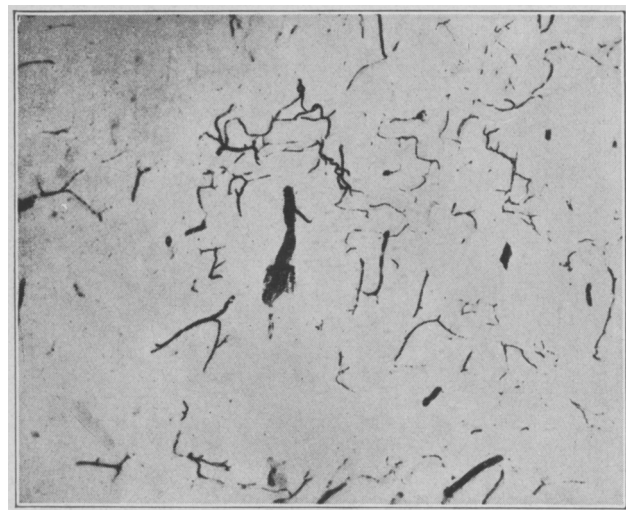


Fig. 8.—Typical distorted vessel surrounded by a ring of congested capillaries from an old plaque of multiple sclerosis. From a brain the blood vessels of which were injected with india ink after removal from the skull. Low power.

sclerosis shows an abnormal tendency to coagulation following the administration of adrenal extract or of typhoid vaccine.

Thrombi are found less often in cases of multiple sclerosis than in cases of "encephalomyelitis," doubtless because most of the lesions are older; but they could be demonstrated in five of fourteen cases.²² Other

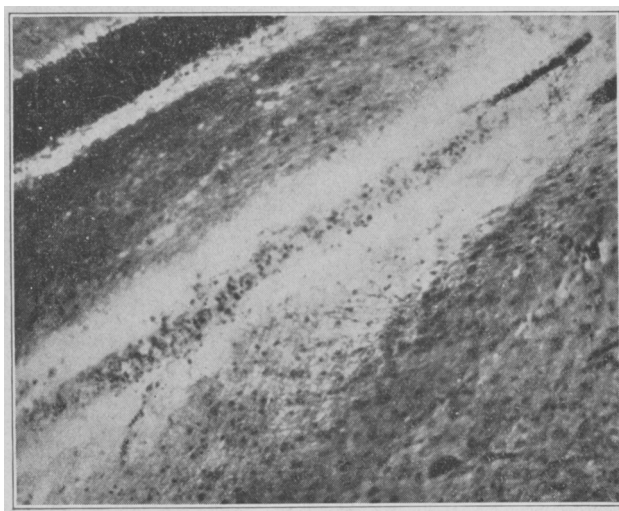


Fig. 9.—Multiple sclerosis. A necrotic vein, devoid of lumen in this and adjacent sections, lies in a zone of demyelination. A congested vein is seen in the upper left corner. Masson stain, high power.

evidences of vascular obstruction, such as perivascular accumulation of blood pigment and engorgement and tortuosity of vessels, are common, however (figs. 8

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